05 - Evidence Pack (Validation Commands & Artifacts)

Document Type: Operational
Phase: Validation
Audience: [engineers, auditors, operations]
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Overview

This document provides **executable commands** to validate current state and capture evidence artifacts for audit/compliance purposes. All commands output JSON or tables suitable for version control and comparison over time.

Output Directory: `i:/eva-foundation/14-az-finops/tools/finops/out/`
Execution Context: Azure CLI authenticated as marco.presta@hrsdc-rhdcc.gc.ca (professional account)
Subscriptions: EsDAICoESub (d2d4e571-e0f2-4f6c-901a-f88f7669bcba), EsPAICoESub (802d84ab-3189-4221-8453-fcc30c8dc8ea)

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1. Storage Account Validation

1.1 List Storage Accounts

Purpose: Enumerate all storage accounts in sandbox resource group.

```
# Command
az storage account list \
  --resource-group EsDAICoE-Sandbox \
  --subscription d2d4e571-e0f2-4f6c-901a-f88f7669bcba \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/storage-accounts.json"

# Expected Output: marcosandboxfinopshub with properties
```

Validation Points:

- [ ] `marcosandboxfinopshub` present in results
- [ ] `kind`: StorageV2 (ADLS Gen2 capable)
- [ ] `isHnsEnabled`: true (hierarchical namespace enabled)
- [ ] `location`: canadacentral
- [ ] `sku.name`: Standard_LRS or Standard_GRS

Evidence Artifact: `storage-accounts.json` (saved to repo)

1.2 List Storage Containers

Purpose: Enumerate all blob containers in FinOps Hub storage.

```
# Command
az storage container list \
  --account-name marcosandboxfinopshub \
  --auth-mode login \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/storage-containers.json"

# Expected Output: Container list with "costs", potentially "raw", "processed", "checkpoint"
```

Validation Points:

- ☐ Container `costs` exists (current location of exports)
- ☐ If migrated: Containers `raw`, `processed`, `archive`, `checkpoint` exist
- ☐ Public access level: None (private)
- ☐ Metadata includes creation timestamp

Evidence Artifact: `storage-containers.json`

Screenshot Required: Portal → Storage Account → Containers blade

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1.3 Sample Blob Listing

Purpose: Validate export file naming and structure.

```
# Command (list last 10 blobs in costs container)
az storage blob list \
  --container-name costs \
  --account-name marcosandboxfinopshub \
  --auth-mode login \
  --num-results 10 \
  --output table

# Expected Output: CSV files with pattern {SubscriptionName}_xxxxxxxx-xxxx.../[date].csv.gz
```

Validation Points:

- ☐ Blob names follow Cost Management naming convention
- ☐ File extension: `.csv.gz` (compressed)
- ☐ Size: >100 KB for daily exports, >1 MB for full exports
- ☐ Last modified: Recent (within 24 hours for daily exports)

Evidence: Copy-paste table output to `storage-sample-blobs.txt`

1.4 Network Rules (Firewall)

Purpose: Verify public access settings.

```
# Command
az storage account show \
  --name marcosandboxfinopshub \
  --resource-group EsDAICoE-Sandbox \
  --query "{publicAccess: publicNetworkAccess, defaultAction:
networkRuleSet.defaultAction, bypass: networkRuleSet.bypass}" \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-
finops/tools/finops/out/storage-network.json"

# Expected Output (before hardening): publicAccess=Enabled, defaultAction=Allow
# Expected Output (after hardening): publicAccess=Disabled, defaultAction=Deny
```

Validation Points:

- ☐ Current state: Public access enabled (development phase)
- ☐ Target state: Public access disabled, VNet rules configured (production phase)

Evidence Artifact: `storage-network.json`

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2. Event Grid Validation

2.1 List Event Grid System Topics

Purpose: Confirm storage account has system topic for blob events.

```
# Command
az eventgrid system-topic list \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-
finops/tools/finops/out/eventgrid-system-topics.json"

# Expected Output: One system topic with source=marcosandboxfinopshub
```

Validation Points:

- ☐ System topic name: `marcosandboxfinopshub-*` (auto-generated)
- ☐ Topic type: Microsoft.Storage.StorageAccounts
- ☐ Provisioning state: Succeeded
- ☐ Source: `/subscriptions/.../resourceGroups/EsDAICoE-Sandbox/providers/Microsoft.Storage/storageAccounts/marcosandboxfinopshub`

Evidence Artifact: `eventgrid-system-topics.json`

2.2 List Event Subscriptions

Purpose: Enumerate event subscriptions (if any) wired to ADF or other endpoints.

```
# Command
az eventgrid system-topic event-subscription list \
  --resource-group EsDAICoE-Sandbox \
  --system-topic-name <system-topic-name-from-above> \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/eventgrid-subscriptions.json"

# Expected Output (current): Empty array (no subscriptions yet)
# Expected Output (future): Subscription to ADF webhook for ingest-costs-to-adx pipeline
```

Validation Points:

- ☐ If empty: No automated ingestion yet (manual validation required)
- ☐ If populated: Endpoint type, filter prefix (`/raw/costs/`), delivery status

Evidence Artifact: `eventgrid-subscriptions.json`

Screenshot Required: Portal → Event Grid System Topic → Metrics (Event Count last 7 days)

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3. Azure Data Factory Validation

3.1 List Data Factories

Purpose: Confirm ADF resource exists in sandbox.

```
# Command
az datafactory list \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/adf-factories.json"

# Expected Output: marco-sandbox-finops-adf
```

Validation Points:

- ☐ Factory name: `marco-sandbox-finops-adf`
- ☐ Location: canadacentral
- ☐ Identity: SystemAssigned or UserAssigned (for authentication)
- ☐ Provisioning state: Succeeded

Evidence Artifact: `adf-factories.json`

3.2 List Pipelines

Purpose: Enumerate configured pipelines (currently unknown).

```
# Command
az datafactory pipeline list \
  --factory-name marco-sandbox-finops-adf \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/adf-pipelines.json"

# Expected Output (current): Empty array OR legacy pipelines
# Expected Output (future): ingest-costs-to-adx, backfill-historical
```

Validation Points:

- ☐ If empty: No automation configured (meets UNKNOWN status from doc 00)
- ☐ If populated: Pipeline activity structure, parameters, triggers

Evidence Artifact: `adf-pipelines.json`

Screenshot Required: Portal → ADF → Author blade showing pipeline list

---

3.3 List Triggers

Purpose: Check if Event Grid triggers are configured.

```
# Command
az datafactory trigger list \
  --factory-name marco-sandbox-finops-adf \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/adf-triggers.json"

# Expected Output (current): Empty or no Event Grid trigger
# Expected Output (future): Storage Event trigger for /raw/costs/ blob creation
```

Validation Points:

- ☐ Trigger type: BlobEventsTrigger
- ☐ Scope: `/subscriptions/.../resourceGroups/.../storageAccounts/marcosandboxfinopshub`
- ☐ Events: Microsoft.Storage.BlobCreated
- ☐ Status: Started

Evidence Artifact: `adf-triggers.json`

3.4 Check Pipeline Run History

Purpose: Validate recent execution (if any).

```
# Command (requires pipeline name, skip if empty)
az datafactory pipeline-run query-by-factory \
  --factory-name marco-sandbox-finops-adf \
  --resource-group EsDAICoE-Sandbox \
  --last-updated-after "2026-02-01T00:00:00Z" \
  --last-updated-before "2026-02-17T23:59:59Z" \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/adf-run-history.json"

# Expected Output (current): Empty array (no runs)
# Expected Output (future): Successful run with duration, row count processed
```

Validation Points:

- ☐ Run status: Succeeded
- ☐ Duration: <10 minutes for single CSV ingestion
- ☐ Output: Row count, target table name

Evidence Artifact: `adf-run-history.json`

---

4. Azure Data Explorer (ADX) Validation

4.1 List ADX Clusters

Purpose: Check if ADX cluster is deployed.

```
# Command
az kusto cluster list \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/adx-clusters.json"

# Expected Output (current): Empty array (NOT deployed yet)
# Expected Output (future): marcofinopsadx, SKU=Dev, state=Running
```

Validation Points (post-deployment):

- ☐ Cluster name: `marcofinopsadx`
- ☐ URI: `https://marcofinopsadx.canadacentral.kusto.windows.net`
- ☐ SKU: Dev_No_SLA_Standard_D11_v2, capacity 2
- ☐ State: Running
- ☐ Provisioning state: Succeeded

Evidence Artifact: `adx-clusters.json`

Note: If empty, this confirms CRITICAL gap from gap analysis (doc 01).

4.2 List ADX Databases

Purpose: Enumerate databases in cluster (post-deployment).

```
# Command (only after cluster deployed)
az kusto database list \
  --cluster-name marcofinopsadx \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/adx-databases.json"

# Expected Output: finopsdb with retention, hot cache settings
```

Validation Points:

- ☐ Database name: `finopsdb`
- ☐ Hot cache period: 31 days
- ☐ Soft delete period: 1825 days (5 years)

Evidence Artifact: `adx-databases.json`

4.3 KQL Validation Queries

Purpose: Validate ADX schema and sample data (post-ingestion).

Commands (execute in ADX Web Explorer: https://dataexplorer.azure.com):

```
// 1. List tables
.show tables

// Expected Output: raw_costs, apim_usage

// 2. Check materialized views
.show materialized-views

// Expected Output: normalized_costs

// 3. Check functions
.show functions

// Expected Output: AllocateCostByApp

// 4. Validate ingestion mappings
.show table raw_costs ingestion csv mappings

// Expected Output: CostExportMapping with 55+ column mappings
```

```
// 5. Row count validation (after ingestion)
raw_costs
| summarize Count=count() by Date
| order by Date desc
| take 30

// Expected Output: Daily row counts matching subscription activity

// 6. Cost trend validation
normalized_costs
| where Date >= ago(30d)
| summarize DailyCost=sum(Cost) by Date, SubscriptionName
| order by Date desc

// Expected Output: Daily cost aggregates matching invoice
```

Evidence: Export query results as CSV, save to `out/adx-validation-queries/`

Screenshot Required: ADX Web Explorer showing table list and sample query results

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5. APIM Validation

5.1 List APIM Instances

Purpose: Confirm APIM resource exists.

```
# Command
az apim list \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/apim-instances.json"

# Expected Output: marco-sandbox-apim
```

Validation Points:

- ☐ Service name: `marco-sandbox-apim`
- ☐ SKU: Developer, Standard, or Premium
- ☐ Location: canadacentral
- ☐ Gateway URL: `https://marco-sandbox-apim.azure-api.net`

Evidence Artifact: `apim-instances.json`

5.2 List APIM APIs

Purpose: Enumerate APIs for policy configuration.

```
# Command
az apim api list \
  --service-name marco-sandbox-apim \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/apim-apis.json"

# Expected Output: List of APIs with display names, paths
```

Validation Points:

- ☐ At least 1 API exists for testing
- ☐ API path, version, protocols (HTTP/HTTPS)

Evidence Artifact: `apim-apis.json`

Screenshot Required: Portal → APIM → APIs blade

5.3 Check Base Policy Configuration

Purpose: Validate if cost attribution headers are injected.

Manual Steps (Azure Portal):

1. Navigate to APIM → APIs → "All APIs"
2. Open "Inbound processing" policy editor
3. Look for `<set-variable>` or `<set-header>` elements extracting:
   - `x-eva-costcenter`
   - `x-eva-caller-app`
   - `x-eva-environment`

Evidence: Export policy XML and save to `out/apim-base-policy.xml`

Validation Points:

- [CURRENT] Policy likely does NOT have these headers (meets UNKNOWN status)
- [TARGET] Policy injects headers with defaults for missing values

---

5.4 Check Application Insights Integration

Purpose: Verify telemetry logging is enabled.

```
# Command
az apim logger list \
  --service-name marco-sandbox-apim \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/apim-loggers.json"
```

```
# Expected Output: Logger linked to App Insights resource
```

Validation Points:

- ☐ Logger type: applicationInsights
- ☐ Resource ID: App Insights instrumentation key or connection string
- ☐ Sampling: 100% (for accurate cost attribution)

Evidence Artifact: `apim-loggers.json`

Screenshot Required: Portal → APIM → Application Insights blade showing diagnostics settings

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6. Cost Management Exports

6.1 List Configured Exports

Purpose: Enumerate active cost exports for both subscriptions.

```
# Command for EsDAICoESub (dev/stage)
az costmanagement export list \
  --scope "/subscriptions/d2d4e571-e0f2-4f6c-901a-f88f7669bcba" \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/cost-exports-dev.json"

# Command for EsPAICoESub (prod)
az costmanagement export list \
  --scope "/subscriptions/802d84ab-3189-4221-8453-fcc30c8dc8ea" \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/cost-exports-prod.json"

# Expected Output: EsDAICoESub-Daily, EsPAICoESub-Daily
```

Validation Points:

- ☐ Export name: `EsDAICoESub-Daily`, `EsPAICoESub-Daily`
- ☐ Schedule: Daily, 1 AM UTC
- ☐ Destination: Storage account `marcosandboxfinopshub`, container `costs`
- ☐ Format: CSV
- ☐ Schema version: 2024-08-01 (ActualCost with tags)
- ☐ Status: Active

Evidence Artifacts: `cost-exports-dev.json`, `cost-exports-prod.json`

Screenshot Required: Portal → Cost Management → Exports showing run history

---

6.2 Validate Export Run History

Purpose: Confirm recent successful executions.

Manual Steps (Azure Portal):

1. Navigate to Cost Management → Exports
2. Select export (e.g., `EsDAICoESub-Daily`)
3. View run history tab
4. Check last 7 days for status (Succeeded) and file size

Evidence: Screenshot showing:

- ☐ Last run date: Within 24 hours
- ☐ Status: Succeeded
- ☐ File size: >1 MB (indicates data)
- ☐ Execution time: <5 minutes

---

7. RBAC and Permissions

7.1 Check Role Assignments on Storage

Purpose: Validate permissions for ADF managed identity (post-deployment).

```
# Command
az role assignment list \
  --scope "/subscriptions/d2d4e571-e0f2-4f6c-901a-
f88f7669bcba/resourceGroups/EsDAICoE-
Sandbox/providers/Microsoft.Storage/storageAccounts/marcosandboxfinopshub" \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-
finops/tools/finops/out/storage-rbac.json"

# Expected Output (future): mi-finops-adf with Storage Blob Data Contributor
```

Validation Points:

- ☐ Principal: Managed identity for ADF (`mi-finops-adf`)
- ☐ Role: Storage Blob Data Contributor
- ☐ Scope: Storage account `marcosandboxfinopshub`

Evidence Artifact: `storage-rbac.json`

7.2 Check Role Assignments on ADX

Purpose: Validate database ingestor role for ADF (post-deployment).

```
// Execute in ADX Web Explorer
.show database finopsdb principals

// Expected Output: mi-finops-adf with role "ingestor"
```

Evidence: Export query result to `out/adx-principals.csv`

7.3 Check Current User Permissions

Purpose: Document user's RBAC roles for audit trail.

```
# Command
az role assignment list \
  --assignee marco.presta@hrsdc-rhdcc.gc.ca \
  --include-inherited \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/user-rbac.json"

# Expected Output: Reader on subscriptions, potentially Contributor on sandbox RG
```

Validation Points:

- ☐ Subscription-level: Reader (minimum)
- ☐ Resource group-level: Contributor (if managing resources)
- ☐ Limitations: Cannot assign roles (requires Owner or User Access Administrator)

Evidence Artifact: `user-rbac.json`

---

8. Monitoring and Diagnostics

8.1 Check Application Insights Availability

Purpose: Confirm App Insights exists for APIM logging.

```
# Command
az monitor app-insights component list \
  --resource-group EsDAICoE-Sandbox \
  --output json | Out-File -FilePath "i:/eva-foundation/14-az-finops/tools/finops/out/appinsights.json"

# Expected Output: App Insights resource (if exists)
```

Validation Points:

- ☐ Component name: `marco-sandbox-appinsights` (or similar)
- ☐ Application type: web
- ☐ Instrumentation key: Non-empty
- ☐ Connection string: Available

Evidence Artifact: `appinsights.json`

8.2 Query Sample Telemetry

Purpose: Validate APIM request logs in App Insights (post-policy deployment).

KQL Query (execute in Portal → App Insights → Logs):

```
requests
| where timestamp > ago(24h)
| where cloud_RoleName == "APIM-marco-sandbox-apim"
| extend CostCenter = tostring(customDimensions["x-eva-costcenter"])
| extend CallerApp = tostring(customDimensions["x-eva-caller-app"])
| summarize RequestCount=count() by CostCenter, CallerApp
| order by RequestCount desc
```

Evidence: Export query result as CSV to `out/appinsights-apim-telemetry.csv`

Validation Points: If query returns no results, cost attribution headers are NOT configured yet.

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9. Summary Checklist

Use this checklist to track evidence collection progress:

Component	Evidence Artifact	Status	Notes
Storage Accounts	`storage-accounts.json`	☐ Not Collected	Run command in section 1.1
Storage Containers	`storage-containers.json`	☐ Not Collected	Run command in section 1.2
Storage Network Rules	`storage-network.json`	☐ Not Collected	Run command in section 1.4
Event Grid Topics	`eventgrid-system-topics.json`	☐ Not Collected	Run command in section 2.1
Event Subscriptions	`eventgrid-subscriptions.json`	☐ Not Collected	Run command in section 2.2
ADF Factories	`adf-factories.json`	☐ Not Collected	Run command in section 3.1
ADF Pipelines	`adf-pipelines.json`	☐ Not Collected	Run command in section 3.2
ADF Triggers	`adf-triggers.json`	☐ Not Collected	Run command in section 3.3
ADX Clusters	`adx-clusters.json`	☐ Not Collected	Run command in section 4.1

Component	Evidence Artifact	Status	Notes
ADX Databases	`adx-databases.json`	☐ Not Collected	Run command in section 4.2 (post-deploy)
APIM Instances	`apim-instances.json`	☐ Not Collected	Run command in section 5.1
APIM APIs	`apim-apis.json`	☐ Not Collected	Run command in section 5.2
APIM Base Policy	`apim-base-policy.xml`	☐ Not Collected	Manual export (section 5.3)
APIM Loggers	`apim-loggers.json`	☐ Not Collected	Run command in section 5.4
Cost Exports (Dev)	`cost-exports-dev.json`	☐ Not Collected	Run command in section 6.1
Cost Exports (Prod)	`cost-exports-prod.json`	☐ Not Collected	Run command in section 6.1
Storage RBAC	`storage-rbac.json`	☐ Not Collected	Run command in section 7.1
User RBAC	`user-rbac.json`	☐ Not Collected	Run command in section 7.3
App Insights	`appinsights.json`	☐ Not Collected	Run command in section 8.1

10. Automation Script

Convenience Script: Execute all CLI commands in batch.

See: `i:/eva-foundation/14-az-finops/tools/finops/az-inventory-finops.ps1`

Usage:

```
cd i:/eva-foundation/14-az-finops/tools/finops
.\az-inventory-finops.ps1

# Output: All JSON files written to out/ directory
# Duration: ~2 minutes
```

Change Log

Date	Author	Change

Date	Author	Change
2026-02-17 08:20 AM ET	Marco Presta	Initial evidence pack with validation commands

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Document Status: Ready for execution
Next Action: Run commands, collect artifacts, generate screenshots